

SDTM, ADaM & TLF

Real-World SAS Practice Questions with Solutions

Ten scenario-based questions modeled on real clinical programming tasks: converting SDTM raw data into ADaM analysis datasets, and building the summary statistics that feed into Tables, Listings, and Figures (TLFs). Each question includes a working SAS solution and an explanation of the logic and why it's written that way.

Q1. SDTM → ADaM Date Conversion

SDTM.DM has a character variable **RFSTDTC** in ISO 8601 format ('2024-01-10'). Derive **TRTSDT** (Date of First Exposure to Treatment) in ADSL as a proper numeric SAS date, displayed as DDMMYYYYYY.

```
data adsl;  
  set dm;  
  TRTSDT = input(RFSTDTC, yymmdd10.);  
  format TRTSDT date9.;  
run;
```

Explanation

The INPUT() function converts a character value into a number using an informat — here, yymmdd10. tells SAS to read a 10-character YYYY-MM-DD string. The FORMAT statement only changes how the value displays (e.g. 10JAN2024); the underlying value is still a SAS date (days since 01JAN1960), which is what lets you do date arithmetic later.

Q2. Age Grouping (AGEGR1)

Derive **AGEGR1** in ADSL using the age bands commonly seen in real protocols: <18, 18-64, and ≥65.

```
data adsl;  
  set adsl;  
  length AGEGR1 $10;  
  if AGE < 18 then AGEGR1 = '<18';  
  else if 18 <= AGE < 65 then AGEGR1 = '18-64';  
  else if AGE >= 65 then AGEGR1 = '>=65';  
run;
```

Explanation

The LENGTH statement is placed before any assignment. Without it, SAS sets the variable's length from whichever value is assigned first in the DATA step compile phase — a common bug where later, longer values get silently truncated. Also note IF/ELSE IF order matters: each condition is only checked if the previous one was false, so ranges don't need to repeat their lower bound.

Q3. Safety Population Flag (SAFFL)

Derive **SAFFL**: 'Y' if the subject actually received any study treatment (i.e. has a non-missing TRTSDT), else 'N'.

```
data adsl;  
  set adsl;  
  length SAFFL $1;  
  if not missing(TRTSDT) then SAFFL = 'Y';  
  else SAFFL = 'N';  
run;
```

Explanation

MISSING() works for both character and numeric variables and is clearer than writing TRTSDT = . directly. Population flags like SAFFL, ITTFL, and EFFFLL are the backbone of ADaM — almost every TLF filters on one of these before summarizing.

Q4. Treatment Duration (TRTDURD)

Derive **TRTDURD**, the number of days a subject was on treatment, given TRTSDT and TRTENDT are already SAS dates.

```
data adsl;  
  set adsl;  
  TRTDURD = TRTENDT - TRTSDT + 1;  
run;
```

Explanation

SAS dates are just integers (days since 1960), so subtracting them gives a day count directly. The +1 makes the count inclusive of both the start and end day — a patient who started and stopped treatment on the same day was on treatment for 1 day, not 0.

Q5. Merging DM and EX for Real Dosing Dates

DM's RFSTDTC/RFENDTC can be unreliable in practice. Derive TRTSDT/TRTENDT instead from SDTM.EX (Exposure), using the subject's first and last actual dosing records.

```
proc sort data=ex; by usubjid exstdtc; run;  
  
data trtdates;  
  set ex;  
  by usubjid;  
  EXSTDTC = input(EXSTDTC, yymmdd10.);  
  retain TRTSDT;  
  if first.usubjid then TRTSDT = EXSTDTC;  
  if last.usubjid then TRTENDT = EXSTDTC;  
  if last.usubjid then output;  
  keep usubjid TRTSDT TRTENDT;  
run;
```

Explanation

FIRST.usubjid / LAST.usubjid only work correctly on data already sorted by that variable — hence the PROC SORT first. RETAIN keeps TRTSDT's value across iterations of the DATA step instead of resetting it to missing every row. Because EX has one row per dose, sorted ascending by date, the first row per subject is their earliest dose and the last row is their most recent.

Q6. Demographic Table Cell Counts (PROC FREQ)

Produce the SEX-by-treatment-arm cross-tabulation that feeds a Table 14.1.1 Demographics table, without SAS cluttering the output with row/column percentages you don't need.

```
proc freq data=adsl;  
  tables TRT01P * SEX / norow nocol nopercnt;  
run;
```

Explanation

By default PROC FREQ prints row %, column %, and cell % for every cross-tab, which is rarely what a demographics table wants. NOROW/NOCOL/NOPERCENT strip those down to just the counts, which is closer to what gets reformatted into the final TLF.

Q7. Continuous Variable Summary (PROC MEANS)

For the same demographics table, summarize AGE (n, mean, std, median, min, max) by treatment arm, rounded to 1 decimal.

```
proc means data=adsl n mean std median min max maxdec=1;
  class TRT01P;
  var AGE;
run;
```

Explanation

CLASS (not BY) is used here because the data doesn't need to be pre-sorted by TRT01P, and PROC MEANS handles the grouping internally. MAXDEC controls decimal display only — it doesn't round the underlying computed statistic.

Q8. Change from Baseline (BDS Structure)

In an ADaM BDS lab dataset (one row per subject/parameter/visit), derive **CHG** (change from baseline) for AVAL, given a baseline record is flagged already.

```
proc sort data=adlb; by usubjid paramcd adt; run;

data adlb;
  set adlb;
  by usubjid paramcd;
  retain BASE;
  if ABLFL = 'Y' then BASE = AVAL;
  CHG = AVAL - BASE;
run;
```

Explanation

This relies on the data being sorted so the baseline record comes before the post-baseline records within each USUBJID/PARAMCD group. RETAIN carries BASE forward from the baseline row so every subsequent row in that group can compute CHG against it. Sort order and RETAIN are doing almost all the real work here.

Q9. Deriving the Baseline Flag Itself (ABLFL)

Before Q8 can work, ABLFL has to be derived: 'Y' for the last non-missing result on or before treatment start date (TRTSDT), per subject/parameter.

```
proc sort data=adlb;
  by usubjid paramcd adt;
run;

data pre_trt;
  set adlb;
  where adt <= trtsdt and not missing(aval);
run;

proc sort data=pre_trt; by usubjid paramcd adt; run;

data ablfl_flagged;
  set pre_trt;
```

```
by usubjid paramcd;  
length ABLFL $1;  
if last.paramcd then ABLFL = 'Y';  
run;
```

Explanation

The WHERE clause first narrows the data down to only candidate baseline records (on/before treatment start, with a real result). Then LAST.paramcd picks the most recent of those per subject/parameter — the standard 'last observation carried as baseline' rule used in most real protocols.

Q10. Demographic Summary Table (PROC TABULATE)

Build the N (column %) layout for SEX by treatment arm that a Table 14.1.1 shell typically expects, in one step.

```
proc tabulate data=adsl;  
  class TRT01P SEX;  
  table SEX, TRT01P * (N COLPCTN);  
run;
```

Explanation

PROC TABULATE's TABLE statement reads as 'rows, columns' — here SEX is the row dimension and TRT01P (crossed with N and COLPCTN) is the column dimension. COLPCTN gives the percentage within each treatment column, which is exactly the '(N=xx) / n (%)' format standard demographic tables use.